Random walks to model a polymer under a force

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1 Background

There are many things that we can model with random walks. One is the configuration of a polymer. We do this by modeling the polymer as a series of sections, the lengths of which are related to the flexibility of the polymer. You can read more here if you are curious. We are even going to simplify this further, by enforcing that each section can only be oriented exactly in the x, y, and z directions, so only 6 possible orientations for individual sections.

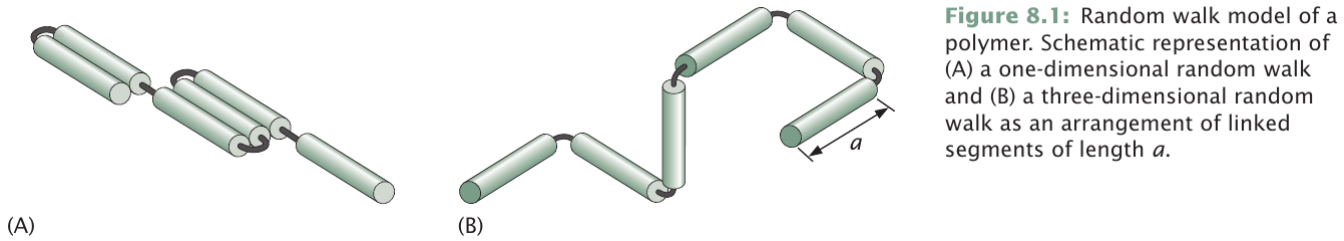


Figure 8.1: Random walk model of a polymer. Schematic representation of (A) a one-dimensional random walk and (B) a three-dimensional random walk as an arrangement of linked segments of length *a*.

Figure 1: An example of one random walk path of a polymer

From the figure, you can probably already see how similar this is to the random walk you've programmed. In fact, that's exactly what it is. So, why are we even talking about this if you've essentially already done this? Well, what if there were a force on the polymer? This would certainly change the configuration on average, so clearly we can't just model it with the same random walk. We can still use a random walk, but we will have to modify it.

2 The problem

We would like to predict the extension of a polymer with a force on it. You may reasonably think that for a suspended polymer any small amount of force should be enough to stretch it into a straight line. However, this is not the case. Because of entropy, the polymer likes to be configured randomly, and it takes a lot more force than you would intuitively think to fully stretch out the polymer. Additionally, there are curves called force extension curves that quantify these relationships, and they are of interest to biophysicists. In fact, one of the members of Poirier Lab does experiments that essentially are just pulling on RNA and seeing what forces you get from certain extensions.

To simplify this, we will model a total force F acting on the ends of the polymer, in the z direction. We will let the pieces of the polymer all be the same length, and either go in the x, y, or z direction, positively or negatively. This is the same setup as previously, but now we will introduce Boltzmann statistics to actually get a different result.

2.1 Getting specific

First, let's define some variables. Let's say we are working with a polymer of length L and segments of length a so that we have a total of N segments, where $N = \frac{L}{a}$. Let us define Z as the extension of the polymer in the z direction. The goal will be to find $\frac{<Z>}{L}$, where $< Z >$ is the average Z over many random walks. Note that $< Z >$ is a very similar measurement to $< R >$, but only in one direction. But wasn't $< R >$ just always zero? Well, for an unbiased random walk that is true, but once we do the physics we will see there will not be an equal chance of moving in every direction.

2.2 Probability theory introduction

To properly understand this problem, we need to understand Boltzmann statistics. To understand Boltzmann statistics (as much as we need to solve this problem), we need to understand some basics of probability theory. To start, here are two core equations of probability:

$$0 \leq P(A) \leq 1$$

$$\sum_{i}^{N} P(A_i) = 1$$

Here, A represents some outcome. For example, for a coin flip, A could represent heads. What the first equation says is that the chance of any outcome cannot be greater than 1 or less than 0. This makes sense, the least likely something can be is 0 and the most likely something can be is 1. The second equation says that for mutually exclusive outcomes (only one outcome A_i can happen, and one must) their probabilities must add to one. This again makes sense, you can just think of this as pulling different colored marbles out of a bag, you must select exactly one, so the probabilities of getting each color added is just the probability of getting a marble at all, which is 1. Now this may seem simple, and it kind of is, but I just want to introduce this language and the ideas of probabilities before moving on.

Questions:

1. Express a single step of a 1-d random walk in terms of outcomes.
2. Verify both of the laws using your representation of a step.
3. Does this work for a 3-d random walk too?

2.3 Boltzmann Statistics

Boltzmann statistics are a classical way to describe probabilities of energy states. Here, we will not go into detail, and rather briefly use it to figure out something about the probabilities to walk each step. Boltzmann statistics' key formula is as such:

$$P(i) = \frac{e^{-E_i/k_B T}}{Z}$$

where

$$Z = \sum_{i} e^{-E_i/k_B T}$$

This is a formula that takes people a while to understand, so don't feel bad if what this formula is saying is not immediately obvious. Here, I am using i as the enumerator of a state, so state i has energy E_i. k_B is the Boltzmann constant (you can look up its value, but until we get to programming this we won't need it) and T is the temperature in Kelvin. This theory says that the probability of a state is proportional to the negative exponential of its energy, and this has important ramifications. You can see that if energy increases, the probability will decrease, and with lower energy, the probability increases. This makes sense, since we expect a system to generally go to its lower energy states.

Questions:

1. Again using the example of a single step of a 1-d random walk, convince yourself that this equation works as expected given the energies are the same for a negative and a positive step.
2. What is the role of Z, called the partition function? What rule is broken if we remove Z from the equation?

2.4 Solving the problem

So, now that we know that we care about energies, how do we describe them for each of the states, or each of the directions of steps? Well, it is actually not very hard. We know that the work done when something moves a distance under a force is $W = dFcos(\theta)$. Recall that the force acts in the Z direction, and use this to find the energy levels for all 6 directions for the random walk.

Now that we know the energy levels, use them to find Z and then the probability of each state.

Finally, we assume we have many random walks, so that $< N_{left} >= N * P(left)$ (and the same for right, down, back, etc.) Write $< Z >$ in terms of $N, N_{left}, N_{right}, N_{up}$, etc, and then put those in terms of probabilities. Use this to write a final equation for $\frac{<Z>}{L}$. You should also significantly simplify your answer by recognizing certain pieces as a hyperbolic cosine and a hyperbolic sine.